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Statistical correlations of shear wave velocity and penetration resistance for soils

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Abstract
In this paper, the correlation between shear wave velocity and standard penetration test blow counts (SPT-*N*) is investigated. The study focused primarily on the correlation of SPT-*N* and shear wave velocity (V_s) for several soil categories: all soils, sand, silt and clay-type soils. New empirical formulae are suggested to correlate SPT-*N* and V_s, based on a dataset collected in a part of Eskişehir settlement in the western central Anatolia region of Turkey. The formulae are based on geotechnical soundings and active and passive seismic experiments. The new and previously suggested formulae showing correlations between uncorrected SPT-*N* and V_s have been compared and evaluated by using the same dataset. The results suggest that better correlations in estimation of V_s are acquired when the uncorrected blow counts are used. The blow count is a major parameter and the soil type has no significant influence on the results. In cohesive soils, the plasticity contents and, in non-cohesive soils except for gravels, the graded contents have no significant effect on the estimation of V_s. The results support most of the conclusions of earlier studies. These practical relationships developed between SPT-*N* and V_s should be used with caution in geotechnical engineering and should be checked against measured V_s.

Keywords: geotechnical soundings, standard penetration test, penetration resistance, shear wave velocity, statistical correlation, Eskişehir, Turkey

Introduction

In geotechnical engineering, many design parameters of soil are associated with the standard penetration test (SPT). SPT is a dynamic *in situ* test, in which a sample tube is driven into the ground to a depth of 45 cm in three successive increments of 15 cm by a 63.5 kg hammer (European Standard is 65 kg) free falling a distance of 76 cm onto an anvil mounted on top of the drill rods. The result quoted is the number of blows (*N*) required to advance the tube for the last 30 cm. SPT-*N* is significant in site investigation, along with other geotechnical parameters such as V_s. Such parameters are accepted as important indicators and are most widely used to describe soil characteristics. It is preferable to determine V_s directly by *in situ* tests, such as by seismic measurements. However, this is not always feasible, due to space constraints and, especially in urban areas, the high noise levels associated with these tests. Therefore, it is necessary to determine V_s through indirect methods such as the SPT test. There is no theoretical relationship between destructive (e.g. SPT) and non-destructive methods (e.g. seismic methods). Therefore, a number of exercises have been carried out with the goal of evaluating the geotechnical properties of soil and of identifying empirical relationships between these properties.

A significant body of research can be found in the literature. Sykora and Koester (1988) found strong statistical correlations between dynamic shear resistance and standard

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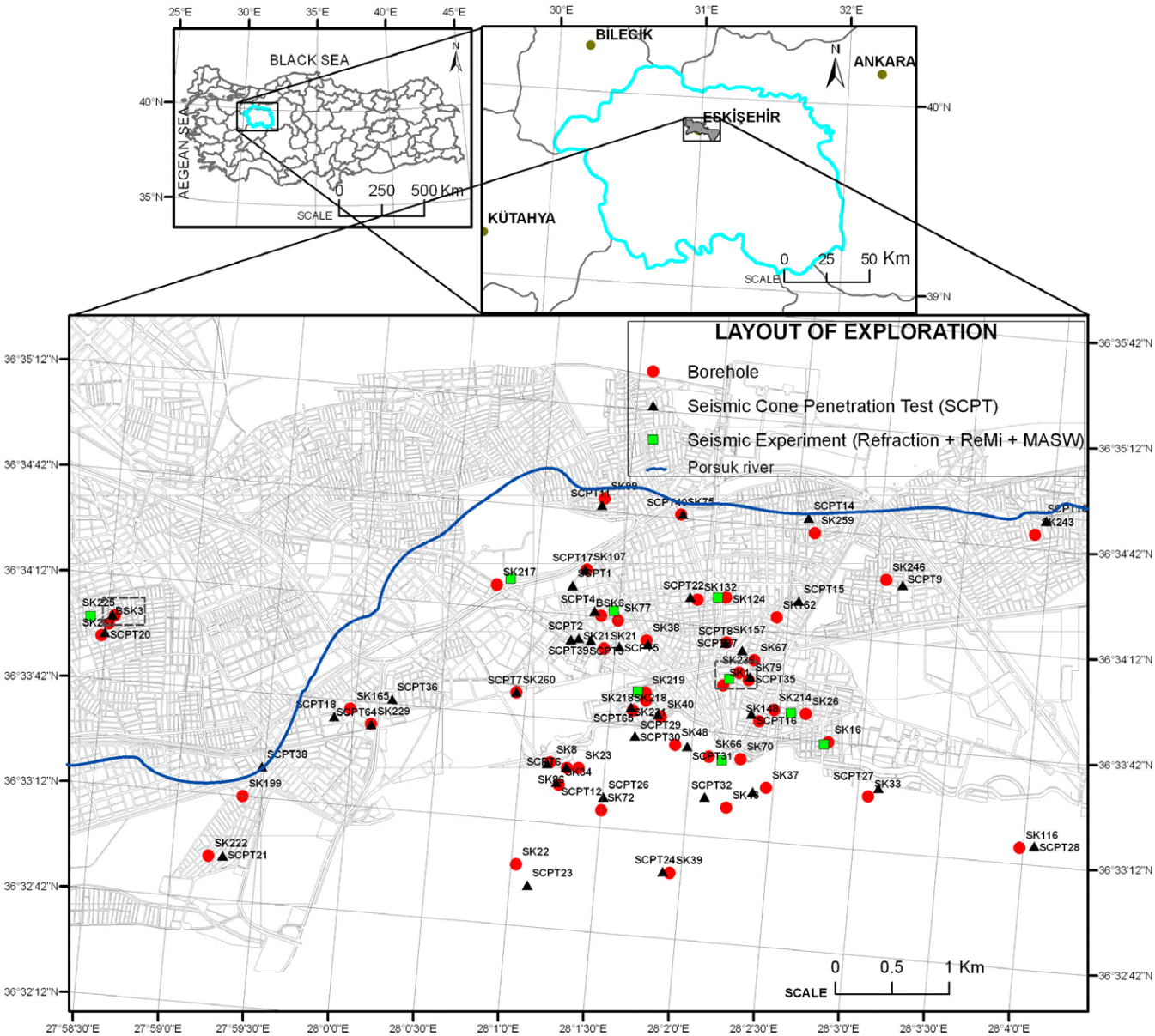

Figure 1. Location of geotechnical and seismic investigation in the study site.

penetration resistance in soils. Jafari *et al* (2002) presented a detailed historical review on the statistical correlation between SPT-N versus V_s. Imai and Yoshimura (1975) studied the relationship between seismic velocities and some index properties over 192 samples and developed empirical relationships for all soils. Sykora and Stokoe (1983) pointed out that geological age and type of soil are not predictive of V_s while the uncorrected SPT-N value is most important. Iyisan (1996) examined the influence of the soil type on SPT-N versus V_s correlation using data collected from an earthquake-prone area in the eastern part of Turkey. The results showed that, except for gravels, the correlation equations developed for all soils, sand and clay yield approximately similar V_s values. Hasancebi and Ulusay (2006) studied similar statistical correlations using 97 data pairs collected from an area in the north-western part of Turkey and developed empirical relationships for all soil types, sand and clay soil types except for gravels. Ulugergerli and Uyanık (2007) investigated statistical correlations using 327 samples collected from different areas of Turkey and defined the empirical relationship as upper and lower bounds instead of a single average curve for estimating seismic velocities and relative density. There are many empirical correlations between SPT-N and V_s in the literature (Shibata 1970, Ohba and Toriuma 1970, Ohta *et al* 1972, Fujiwara 1972, Ohsaki and Iwasaki 1973, Imai and Yoshimura 1975, Campbell and Duke 1976, Imai 1977, Ohta and Goto 1978, Seed and Idriss 1981, Imai and Tonouchi 1982, Barrow and Stokoe 1983, Jinan 1987, Okamoto *et al* 1989, Lee 1990, Athanasopoulos 1995, Sisman 1995, Kanai 1966, Jafari *et al* 1997, Pitilakis *et al* 1999, Kiku *et al* 2001, Tamura and Yamazaki 2002). Some researchers have proposed correlations between SPT-N and V_s for different soils, such as sand, silt and clay. Others have developed correlation equations which included stress-

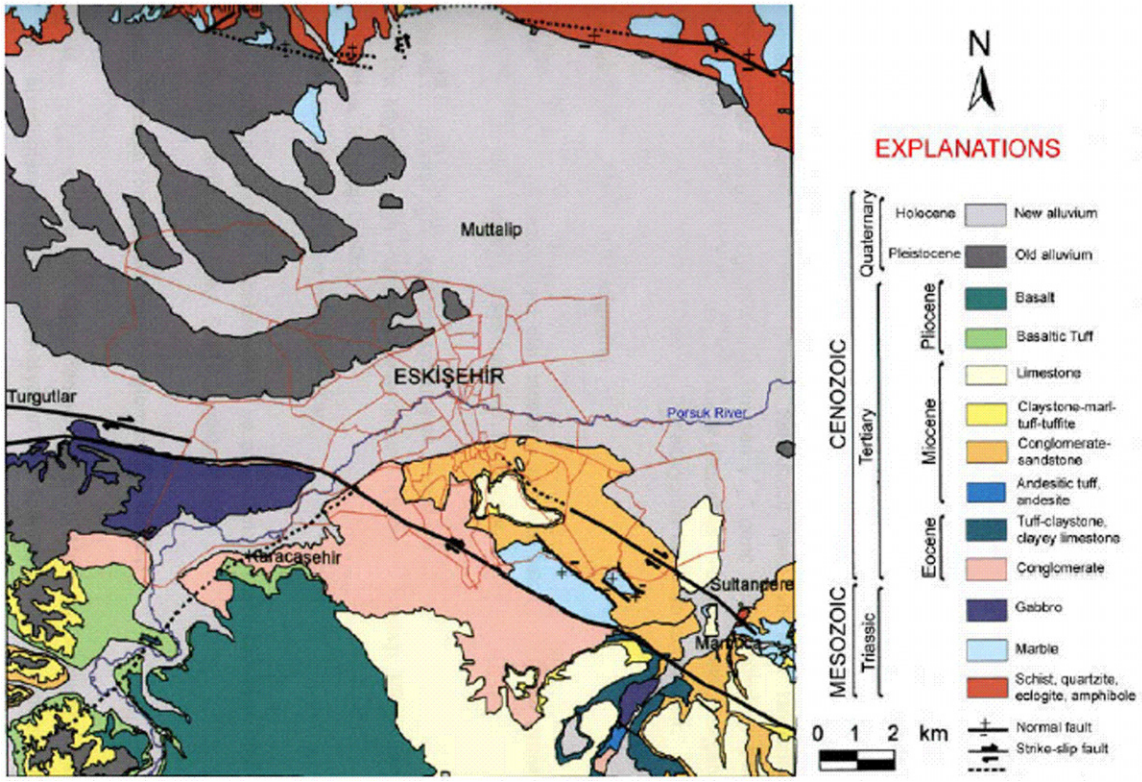

Figure 2. Geological map of the study site (Ayday *et al* 2001).

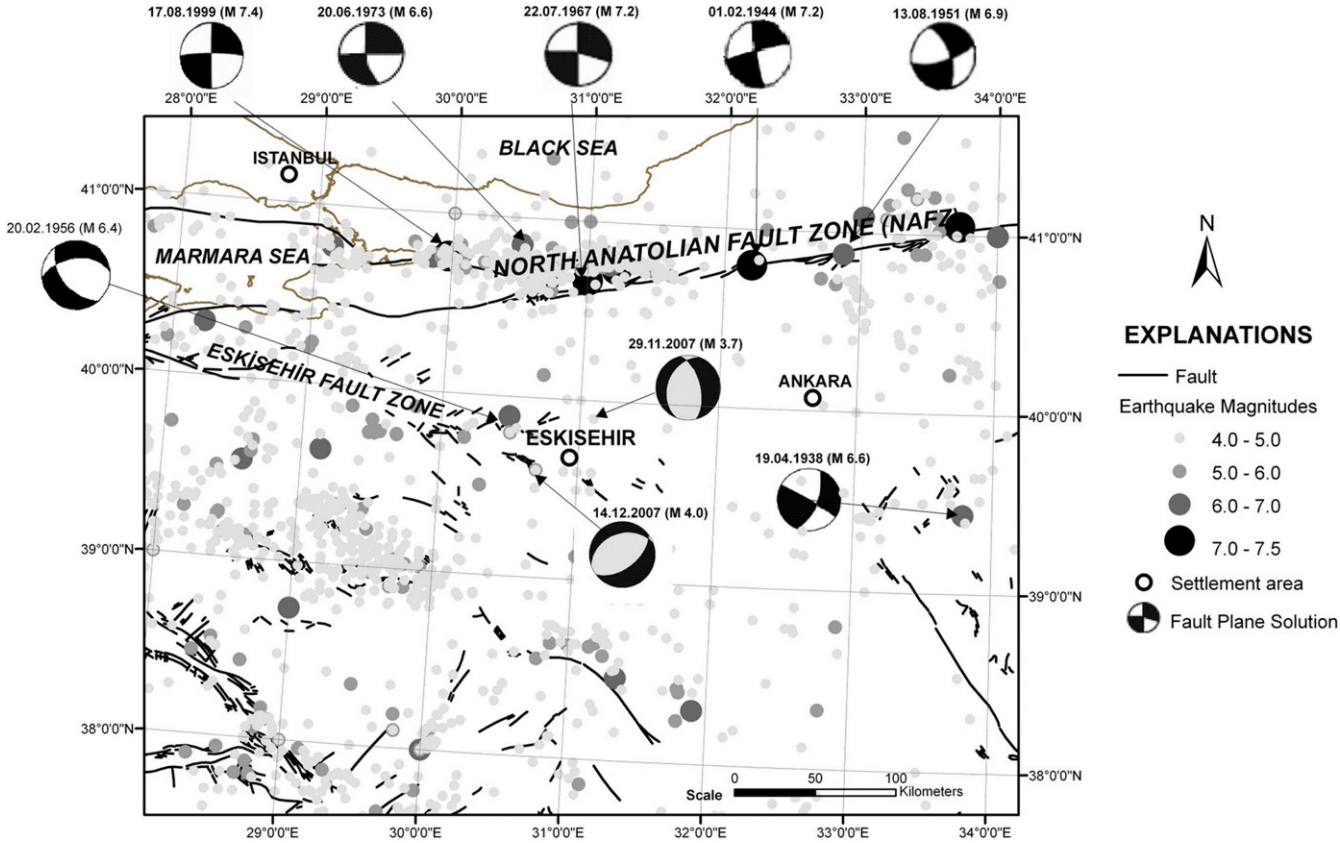

Figure 3. Seismotectonics map of the study site and surroundings.

corrected V_s, energy-corrected SPT-N (e.g. Pitilakis *et al* 1999), energy- and stress-corrected SPT-N, depth (e.g. Lee 1990, Tamura and Yamazaki 2002) and fine content (e.g. Imai 1977, Ohta and Goto 1978, Okamoto *et al* 1989). However, with the

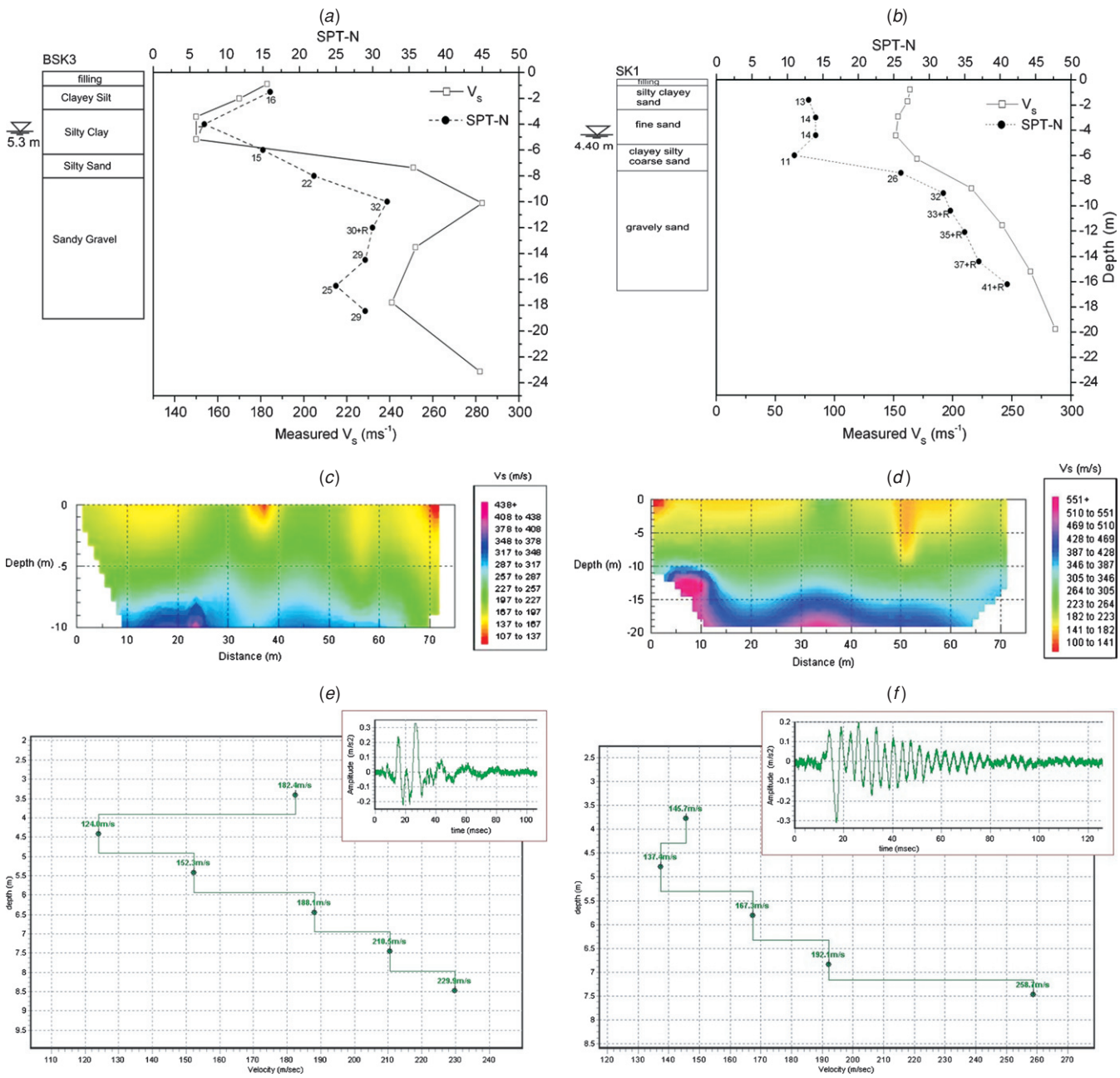

Figure 4. (*a*), (*b*) Some typical SPT-*N* and V_s variation with depth; (*c*), (*d*) seismic refraction profiles; (*e*), (*f*) V_s profiles obtained from the SCPT at the study site.

exception of Lee (1990), almost all of the studies mentioned above focused on the relationships between uncorrected SPT-*N* and V_s for all soils as well as sand and clay-type soils.

In the present study, the statistical correlation between uncorrected SPT-*N* and V_s was investigated for all soils, sand, silt and clay-type soils. A new empirical correlation equation is proposed to estimate V_s directly from uncorrected SPT-*N* values using statistical analysis for all soils, sand, silt and clay-type soils. To investigate predictive capability, these correlation equations are compared with the previously suggested equations. A part of Eskişehir settlement founded on an alluvial plain was selected as the study site (figure 1). Eskişehir is one of the industrialized cities located in the western central part of Turkey and has a rapidly expanding population. The field work included SPT borings, cone penetration tests (CPTs), seismic cone penetration tests (SCPTs) and seismic studies, namely refraction microtremor (ReMi), multi-station analysis of surface waves (MASW) and refraction seismic methods. The rest of the study consisted of laboratory tests, borehole data from previous research at the study site and statistical analysis.

General setting of the study site

Geological and seismotectonic setting

The geological map of the study area is shown in figure 2. A considerable part of the city of Eskişehir is founded

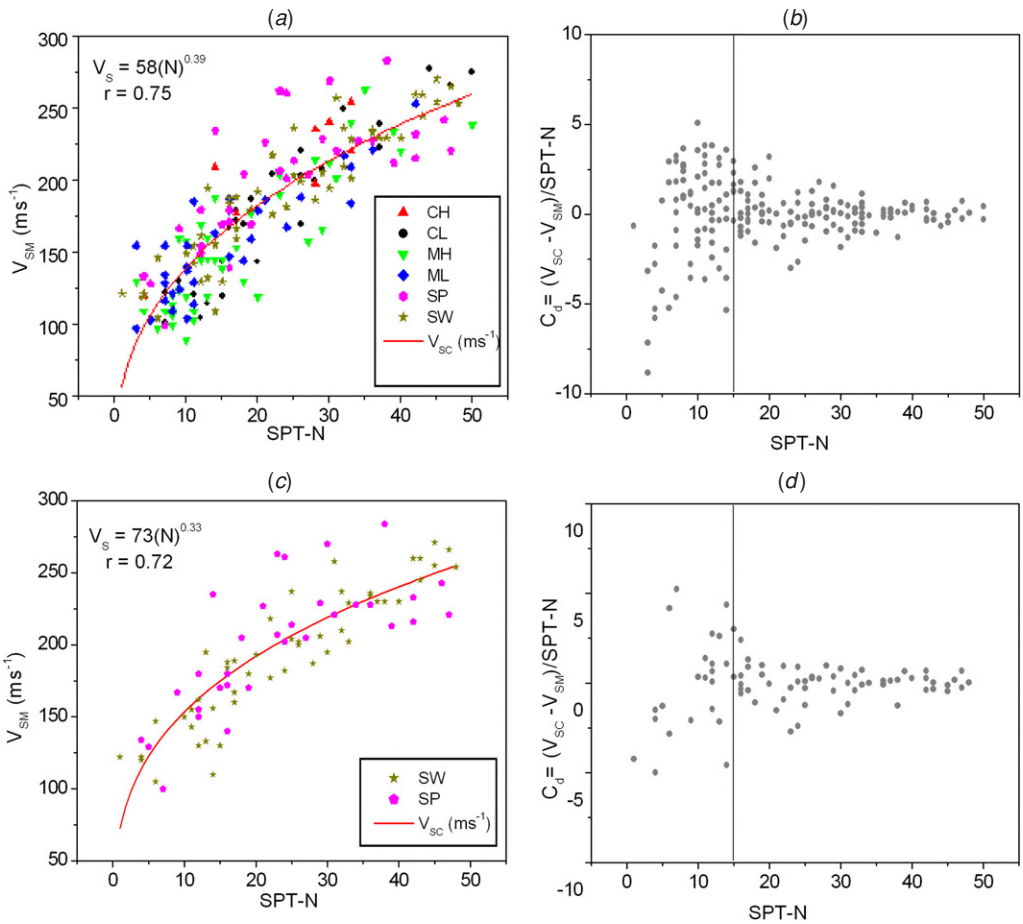

Figure 5. Correlations between SPT-N and V_s values: (a) for all soils, (b) normalized consistency ratio for all soils, (c) for sand soils, (d) normalized consistency ratio for sand soils, (e) for silt soils, (f) normalized consistency ratio for silt soils, (g) for clay soils and (h) normalized consistency ratio for clay soils.

Table 1. Grain size distributions of soils from the study site.

Grain size	Min	Max	Mean	Standard error	Standard deviation
Gravel (%)	0	85	13.96	21.05	0.030
Sand (%)	2	83	26.66	16.30	0.023
Silt (%)	0	78	36.54	19.79	0.028
Clay (%)	0	71	22.84	15.18	0.021

on quaternary alluviums. Three lithological units were distinguished by Ayday *et al* (2001) in the settlement area of Eskişehir. These units are (a) Conglomerate member of the Lower Eocene Mamuca Formation, (b) Conglomerate-sandstone, claystone-marl-tuff-tuffite and limestone members of the Upper Miocene Porsuk Formation and (c) old and recent quaternary alluviums.

Based on the earthquake zonation map of Turkey (General Directorate of Disaster Affairs—GDDA 1996), Eskişehir is situated within the second degree earthquake region and located between different fault systems defined by distinct fault characteristics with respect to each other. The Eskişehir Fault Zone (EFZ) and North Anatolian Fault Zone (NAFZ) are the fault zones nearest to Eskişehir city (figure 3). The city has been affected by past earthquakes (e.g. the 1999 Izmit earthquake $M_w = 7.4$) and a number of buildings collapsed.

SPT soundings, CPT, SCPT, seismic investigations and laboratory testing

The dataset used in this study consists of three main sources. In order to determine the conditions and characteristics of the soils in the study site, SPT boreholes ranging in depth from 4.5 m to 30.45 m were drilled at 264 different locations using a D-200 model drilling rig (Polmak Corp.). Additionally, CPT, ranging in depth from 4 m to 15 m, was conducted at 45 different locations. These tests were carried out by the Civil Engineering Department of Anadolu University, Turkey, in the summer of 2000 and 2001. The SPT in all boreholes was performed using the following steps. (i) A standard split-barrel sampler was used. (ii) The sampler was driven into the ground to various depths by a 63.5 kg slide-hammer free falling from a height of 76 cm onto an anvil mounted on top of the drill rod. (iii) The number of blows required to advance the sampler for the last 30 cm was quoted. SPTs were carried out from boreholes at different depths, varying between 1 m and 3 m. The groundwater table in each borehole was also measured and generally varied between 3 and 12 m across the

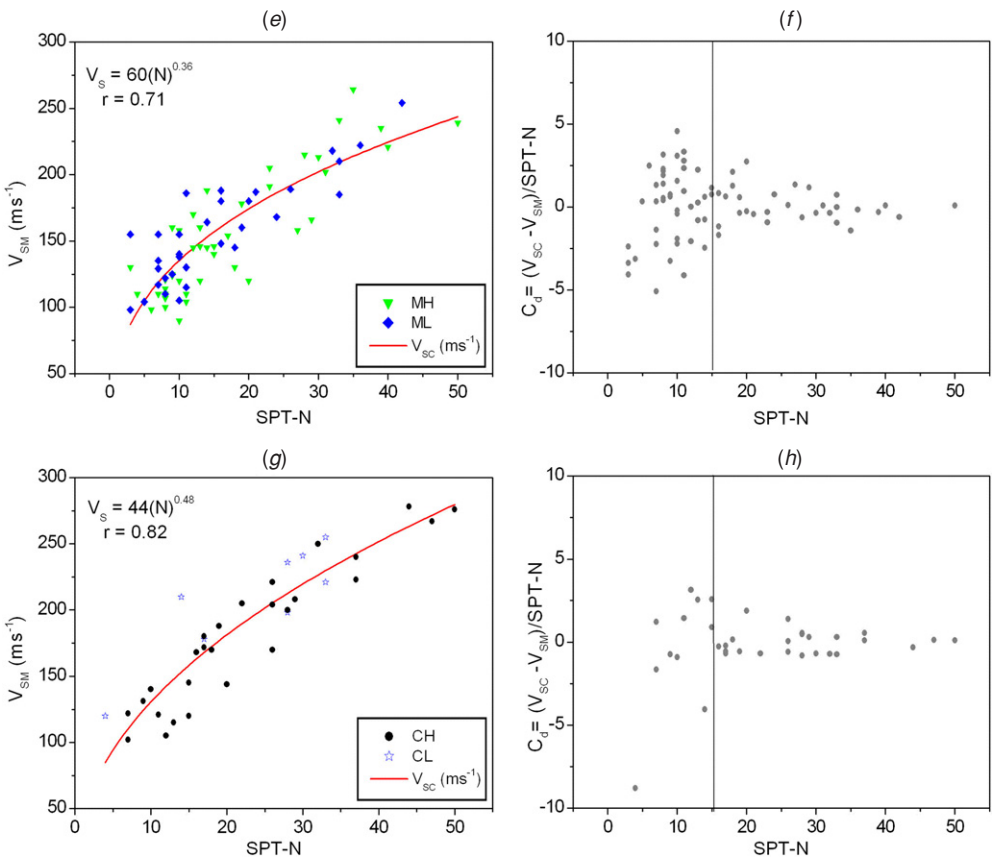

Figure 5. (Continued.)

Table 2. Some existing correlations between uncorrected SPT-N and V_s.

	Author(s)	All soils	Sand	Silt	Clay
1	Shibata (1970)	–	$V_s = 31.7\ N^{0.54}$		–
2	Ohba and Toriuma (1970)	$V_s = 84\ N^{0.31}$	–		–
3	Imai and Yoshimura (1975)	$V_s = 76\ N^{0.33}$	–		–
4	Ohta *et al* (1972)	–	$V_s = 87.2\ N^{0.36}$		–
5	Fujiwara (1972)	$V_s = 92.1\ N^{0.337}$	–		–
6	Ohsaki and Iwasaki (1973)	$V_s = 81.4\ N^{0.39}$	–		–
7	Imai *et al* (1975)	$V_s = 89.9\ N^{0.341}$	–		–
8	Imai (1977)	$V_s = 91\ N^{0.337}$	$V_s = 80.6\ N^{0.331}$		$V_s = 80.2\ N^{0.292}$
9	Ohta and Goto (1978)	$V_s = 85.35\ N^{0.348}$	–		–
10	Seed and Idriss (1981)	$V_s = 61.4\ N^{0.5}$	–		–
11	Imai and Tonouchi (1982)	$V_s = 97\ N^{0.314}$	–		–
12	Sykora and Stokoe (1983)	–	$V_s = 100.5\ N^{0.29}$		–
13	Jinan (1987)	$V_s = 116.1(N{+}0.3185)^{0.202}$	–		–
14	Okamoto *et al* (1989)	–	$V_s = 125\ N^{0.3}$		–
15	Lee (1990)	–	$V_s = 57.4\ N^{0.49}$	$V_s = 105.64\ N^{0.32}$	$V_s = 114.43\ N^{0.31}$
16	Athanasopoulos (1995)	$V_s = 107.6\ N^{0.36}$	–		$V_s = 76.55\ N^{0.445}$
17	Sisman (1995)	$V_s = 32.8\ N^{0.51}$	–		–
18	Iyisan (1996)	$V_s = 51.5\ N^{0.516}$	–		–
19	Kanai (1966)	$V_s = 19\ N^{0.6}$	–		–
20	Jafari *et al* (1997)	$V_s = 22\ N^{0.85}$	–		–
21	Kiku *et al* (2001)	$V_s = 68.3\ N^{0.292}$	–		–
22	Jafari *et al* (2002)	–	–		$V_s = 27\ N^{0.73}$
23	Hasancebi and Ulusay (2006)	$V_s = 90\ N^{0.309}$	$V_s = 90.82\ N^{0.319}$		$V_s = 97.89\ N^{0.269}$
24	Ulugergerli and Uyanık (2007)	[a]$V_{SU} = 23.291\mathrm{Ln}(N)+405.61$ [b]$V_{SL} = 52.9\,\mathrm{e}^{-0.011N}$	–		–

[a] Upper bound.
[b] Lower bound.

study area. The SCPT at the CPT locations was performed by pushing an instrumented cone-tipped rod into the ground at a constant rate using a modified drilling rig, and V_s was recorded digitally at intervals of 1 m. During the CPT tests the tip resistance, sleeve friction and dynamic pore pressure were recorded digitally to determine stratigraphy. The remaining borehole data (SPT-*N* values) were obtained from geotechnical reports carried out by companies operating in the study site. Seismic studies including ReMi, MASW and seismic refraction methods were performed at nine locations by the Geophysical Engineering Team of Ankara University, Turkey (Başokur *et al* 2008) to evaluate the shear wave velocity profile in the study site. The locations of these boreholes, SCPTs and seismic measurements are shown in figure 1. Disturbed and undisturbed samples (700 in total) were collected from the boreholes and tested in the laboratory of Hacettepe University, Turkey. Information about the soil classification, fine content, water content, unit weight, sieve analysis and Atterberg limits were obtained from the laboratory tests.

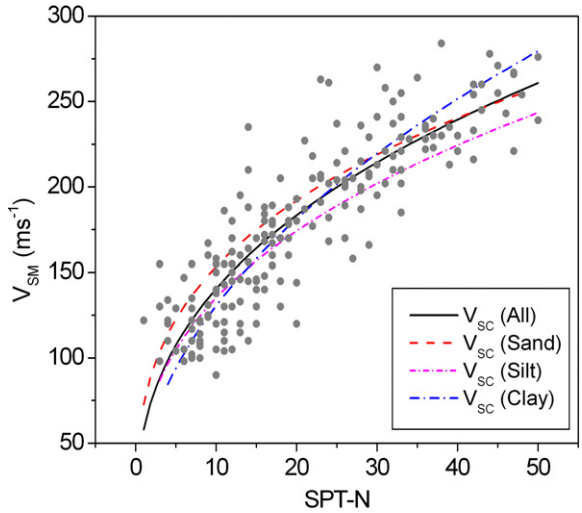

Figure 6. Effect of the soil type on SPT-*N* versus V_s.

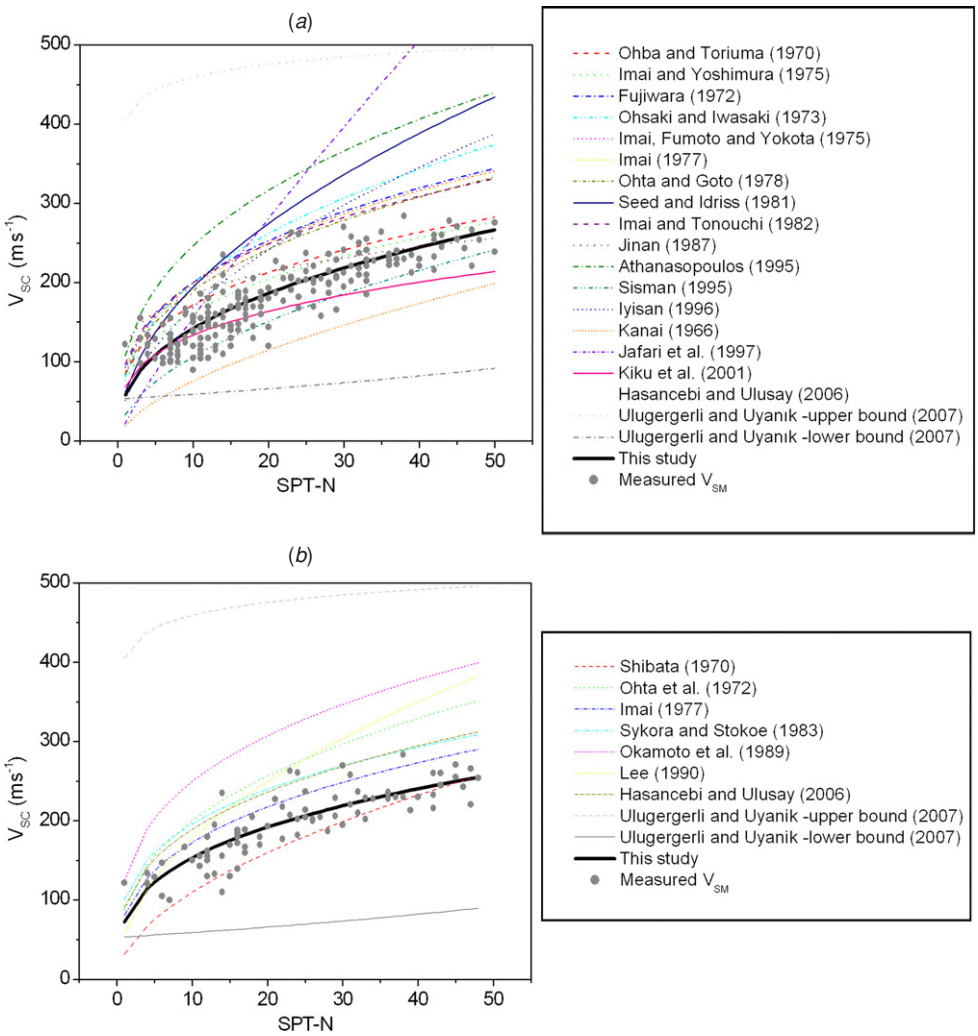

Figure 7. Comparisons between proposed and previous correlations for SPT-*N* and V_s: (*a*) for all soils, (*b*) sand soils (*c*) silt soils and (*d*) clay soils.

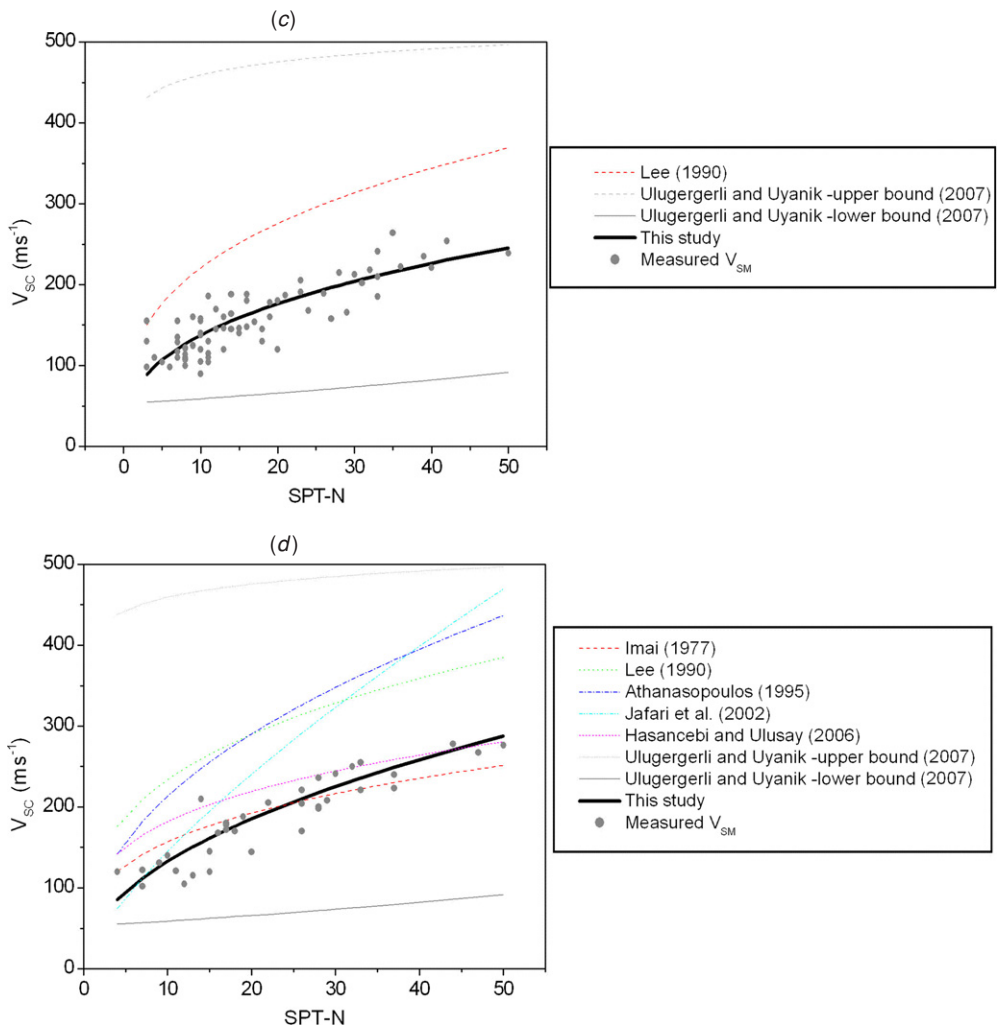

Figure 7. (Continued.)

Subsurface conditions

The data obtained from previous research and recent geotechnical studies indicate that the site is mostly composed of alluvial and rock units. Based on the available information, there are two different major alluvial units which can be defined as old and recent alluvial deposits. The north-west part of the study site is composed of old alluvial deposits and the southern part is covered by rock units. The records of earthquakes indicate that the recent alluvial deposits show high risk in the site. Recent alluvial deposits consist of loose sediments, and the thickness of organic soil at the upper level of this unit varies occasionally (Ayday *et al* 2001). Below this level, a silt-sand unit and a thick clay layer can be observed at some regions. Areas below this level consist of sandy and a pebble-sand material.

In order to determine the physical properties of soil samples obtained from SPT borings, laboratory tests including sieve analysis, Atterberg limit analysis, water content analysis, unit weight analysis, and triaxial shearing test and consolidation tests were accomplished. Laboratory tests of 700 samples reveal that the unit weight distribution of soils in the study site varies between 1.86 g cm^{-3} and 2.0 g cm^{-3} for gravel, 1.90 g cm^{-3} and 2.1 g cm^{-3} for sand and 1.81 g cm^{-3} and 2.0 g cm^{-3} for silt and clay. Results from the sieve analysis and the statistical grain size distribution according to the Unified Soil Classification System (USCS) (ASTM D-2487 2000) are given in table 1. This indicates predominantly silt-sand units and limited clay and gravel deposits. The laboratory tests show that the clay layers of alluvial units have both low and high plasticity (ML-MH) and contain mica grains and inorganic silt with fine sand. According to the information obtained from seismic experiments, V_s of the uppermost 2 m is usually low (120–180 m s^{-1}) in the alluvial site. All seismic profiles agree well with the borehole data and indicate that the soils in the study site display increasing stiffness with depth. The significant decrease in V_s at an average depth of about 5 m indicates the location of the groundwater table. Some selected geotechnical logs and two typical seismic profiles obtained by ReMi and MASW experiments showing the variation of V_s at the two boreholes are depicted in figures 4(*a*) and (*b*); seismic

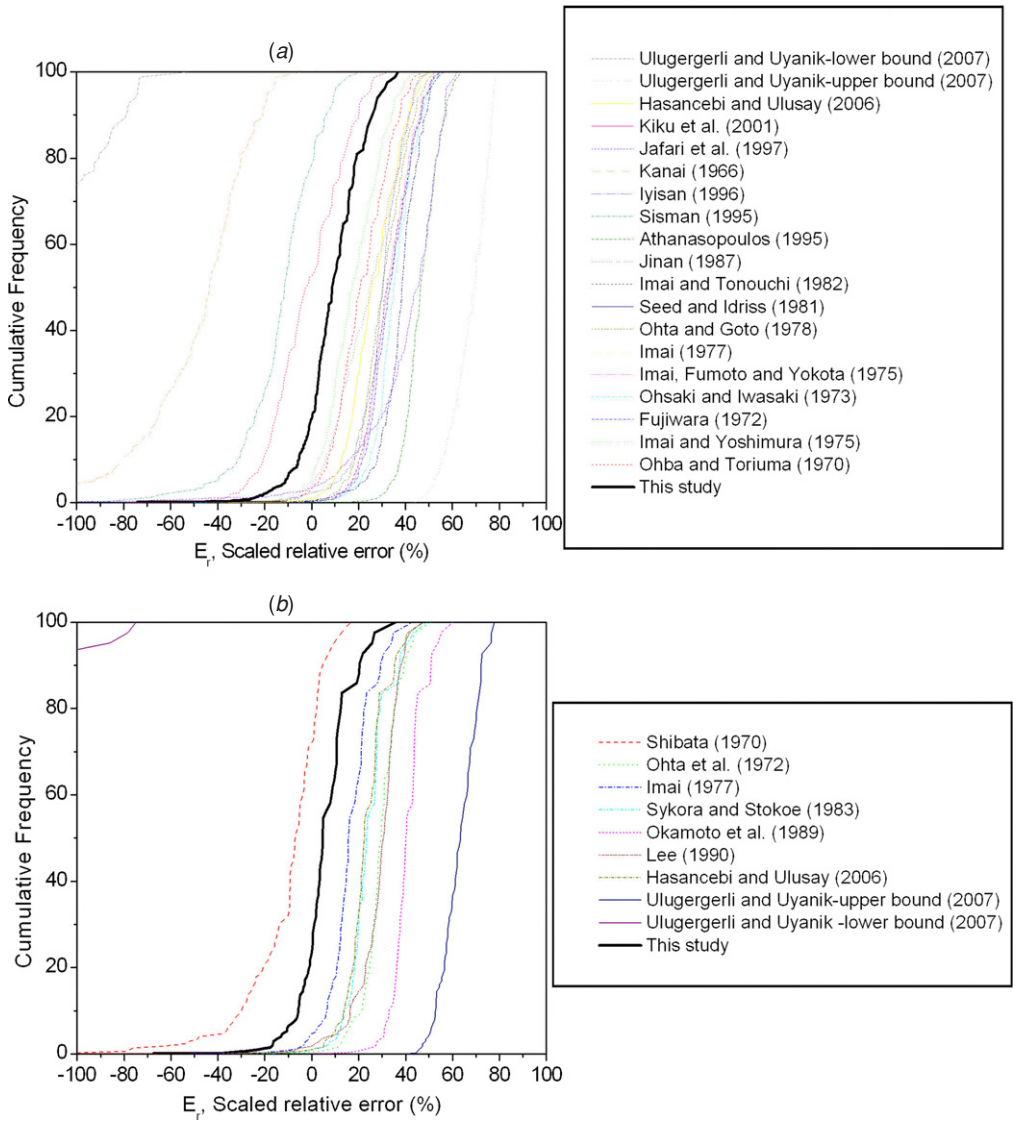

Figure 8. Scaled relative errors of V_s predicted for (*a*) all soils, (*b*) sand soils, (*c*) silt soils and (*d*) clay soils. (This figure is in colour only in the electronic version)

refraction profiles and the SCPT for the same locations are shown in figures 4(*c*) and (*d*) and (*e*) and (*f*), respectively. The location of the sampling points is indicated by the rectangles in figure 1.

Proposed empirical correlations between SPT-*N* and V_s

The published literature contains many equations describing the correlation between SPT-*N* and V_s. Some are material dependent (sand, silt and clay), while others depend on depth, fine contents or corrected penetration resistance $(N_1)_{60}$ and geological age. The previously published empirical formulae that describe the relationship between uncorrected SPT-*N* and V_s are shown in table 2. Evaluation of some of these published relationships revealed that most did not match well against the local data in the present study. However, with the exception of Lee (1990), almost all studies focused on the relationships between uncorrected SPT-*N* and V_s for all soils, sand and clay-type soils.

In the present study, 193 uncorrected SPT-*N* and V_s data pairs consisting of 82 sand, 76 silt and 35 clay samples were obtained from 52 boreholes, 43 SCPT tests and 9 seismic experiments. In statistical analysis, all data were separated according to high or low plasticity for cohesive soils and uniform or poor gradation for sand soil according to the results of laboratory tests. SPT-*N* values used in statistical analysis were obtained from different depths, ranging from 3 m to 30.45 m. The penetration depth of seismic waves sweeps the entire site; hence, the depth information of SPT is ignored in the correlations, except for samples taken in levels corresponding to seismic layers. As a first step, statistical correlations with their correlation coefficients (*r*) between uncorrected SPT-*N* and corresponding V_s values for all soils, sand, silt and clay soils have been obtained using nonlinear regression. The method is based on the Levenberg–Marquardt

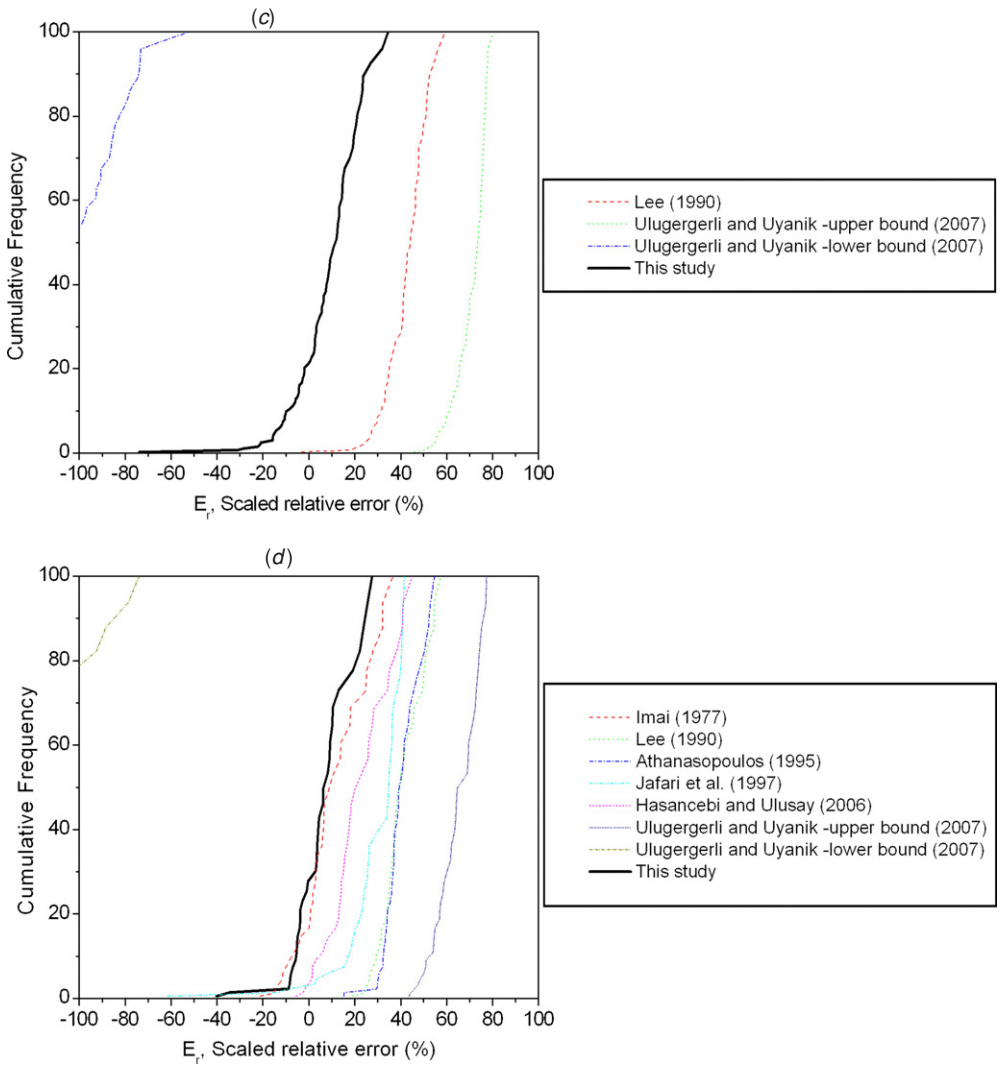

Figure 8. (Continued.)

algorithm, the most widely used nonlinear algorithm in least squares analysis. The following empirical formulae were obtained by using the existing dataset:

$$V_s = 58N^{0.39} \quad (r = 0.75 \text{ for all soils}), \tag{1}$$

$$V_s = 73N^{0.33} \quad (r = 0.72 \text{ for sand soil}), \tag{2}$$

$$V_s = 60N^{0.36} \quad (r = 0.71 \text{ for silt soil}), \tag{3}$$

$$V_s = 44N^{0.48} \quad (r = 0.82 \text{ for clay soil}). \tag{4}$$

The dataset and the fitted curves for the above formulae are shown in figures 5(*a*), (*c*), (*e*) and (*g*), respectively. High correlation coefficients in the formulae produced indicate that the SPT-*N* value has a major effect in V_s estimation. It can be seen that SPT-*N* values obtained from different types of soil including high or low plasticity and uniform or poor gradation are randomly distributed (figure 5). This shows that the type of soil has no significant effect on estimation of V_s. This result is also consistent with the findings of Sykora and Stokoe (1983), Sisman (1995), Iyisan (1996) and Hasancebi and Ulusay (2006). The normalized consistency ratio, C_d, is given as

$$C_d = (V_{SM} - V_{SC})/\text{SPT}{-}N, \tag{5}$$

where V_{SM} is measured V_s from the SCPT and seismic experiment, V_{SC} is calculated V_s from equations (1)–(4) and SPT-*N* is uncorrected SPT blow counts corresponding to V_{SM}. Comparison between V_{SM} and V_{SC} to assess the predictive capability of the equations is shown in figures 5(*b*), (*d*), (*f*) and (*h*). C_d values fall close to zero which means that the proposed equations have good performance in prediction of V_s, except for small SPT-*N* values (SPT-$N < 15$). The depths of small SPT-*N* values (SPT-$N < 15$) range from 4.5 m to 28 m. Therefore, the depth may not be considered as an effective parameter on correlation.

All the fitted curves for different types of soils, sand, silt and clay are plotted in the same figure to evaluate the effect of the soil type (figure 6). Figure 6 implies that the correlations for different soil types yield similar V_s values which mean that the soil type has little effect on these correlations. This is consistent with the earlier studies of Iyisan (1996) and Hasancebi and Ulusay (2006). The V_{SC} values calculated by using formulae produced in this paper and the previously suggested formulae given in table 2 versus uncorrected SPT-*N* values are plotted for different types of soils in figure 7. Athanasopoulos (1995), Seed and Idriss (1981), Fujiwara

(1972), Ohsaki and Iwasaki (1973), and Iyisan (1996) give high V_s values, and these differences increase with increasing SPT-N value for all soils (figure 7(a)). Kanai (1966) and Kiku *et al* (2001) give lower V_s values for all soils. The upper bound method suggested by Ulugergerli and Uyanık (2007) gives much higher V_s values while the lower bound shows a weak approximation. All the other correlations given in table 2 show minor differences and give similar V_s values for all soils. Similar comparisons are made for sand soil and depicted in figure 7(b). The relationships presented by Okamoto *et al* (1989), Lee (1990) and Ohta *et al* (1972) predict higher V_s values while the others, except that by Shibata (1970), which predict lower V_s values, show little difference. However, this difference increases with increasing SPT-N values for sand soils. Ulugergerli and Uyanık (2007) give much higher V_s values and lower V_s values for lower bound approximation. The comparison for silt soil given in figure 7(c) reveals that the formulation by Lee (1990) predicts higher V_s values. For silt soil, Ulugergerli and Uyanık (2007) give much higher V_s values for upper bound approximation and lower values for lower bound approximation. The comparisons for clay soil displayed in figure 7(d) show that Athanasopoulos (1995), Lee (1990) and Jafari *et al* (2002) give higher V_s values. Hasancebi and Ulusay (2006) and Imai (1977) show little difference. Ulugergerli and Uyanık (2007) give much higher V_s values and lower V_s values for lower bound approximation for clay-type soil.

To gain an insight into the capabilities of the proposed correlations, the relative error, E_r, in per cent, is given by

$$E_r = 100(V_{SC} - V_{SM})/V_{SC}. \quad (6)$$

As seen in figure 8(a), using relationship (1) for all soils, about 90% of the V_s values were predicted within a 20% error margin. Using equation (2), 98% of the V_s values were predicted within 20% error for sand soil (figure 8(b)). For silt soils, 90% of the V_s values were predicted within 18% error (figure 8(c)) and for clay-type soils, about 90% of the V_s values were predicted within 20% error (figure 8(d)). These results show that the proposed relationships for all soils, sand, silt and clay-type soils give a better estimation than those from previous existing correlations. However, all of the correlation equations obtained in this study are close to most of the other previously published results. Differences have been seen between existing and proposed correlations. The reason for the differences may be due to specific geotechnical conditions of the study area, geological age, over-consolidation or water table fluctuations affecting correlations considerably. In addition, the variability of SPT equipment and procedures used has significant effects on the blow counts obtained (Seed and Idriss 1981, Iyisan 1996, Jafari *et al* 2002). For example, the energy delivered to the split-spoon sampler is strongly influenced by many factors such as the type of hammer release equipment, diameter of the rope, hammer type, borehole diameter, rod length and rod diameter, verticality of the rod string and even expertise of the operator. Different methods of shear wave velocity measurements and the usage of a special dataset may also be causes of the differences observed. Therefore, different correlation equations can be expected between existing correlations and proposed in this study for the same type of soil.

Conclusions

In summary, the study location was located in a part of Eskişehir settlement in the western central part of Turkey. Data were collected from 52 boreholes, 43 SCPTs, geophysical surveys and geotechnical reports. Data were analysed statistically and compared with previous results within the literature. In this study, an attempt was made to develop new relationships between uncorrected SPT-N and V_s, which is the most important parameter for soil characterization to be used for practical purposes in geotechnical engineering. The results obtained in the present study reveal that the uncorrected blow count has a major effect in the estimation of V_s. On the other hand, some researchers in the literature, such as Hasancebi and Ulusay (2006), used energy-corrected SPT-N values in correlation estimation. However, their findings show a low correlation coefficient. The plasticity contents for cohesive soils and the graded contents for non-cohesive soils, except for gravels, have no significant effect on V_s estimation. The soil type does not significantly affect the correlation between uncorrected SPT-N and V_s. Investigation of previous correlations between SPT-N and V_s showed that previous researchers used soils with different physical properties, for example fine content, water content, pore ratio, unit weight, etc; therefore, different relationships can be expected between existing correlations and those proposed in this study. All the results obtained from this study and previous research reveal that empirical correlations derived from a local dataset should not be used to approximate V_s directly from SPT-N values for different sites. Therefore, these proposed relationships should be used with caution in geotechnical engineering and should be checked against measured V_s.

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